LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	22
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Response of servomechanism and damping and its types
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Response of servomechanism and damping and its types.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of servomechanism response curves; Safety: Ensure electrical isolation; Supervision: Instructor monitors equipment operation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an AFCS?
(b) Define a servomechanism in the context of flight control.
(c) Why is damping necessary in a control system?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of servomechanism response • Transient response characteristics • Steady state error analysis • Importance of stability criteria	L & Ds	PPT, IB	15 min
(b)	• Underdamped system behavior • Overdamped system behavior • Critically damped system behavior • Effect of damping ratio on response	L & Ds	PPT, PDF	20 min
(c)	• Practical implications of damping types • Selection criteria for AFCS damping • Troubleshooting damping issues • Maintenance checks for servomechanisms	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between underdamped and overdamped responses.
(b) How does damping affect the stability of an AFCS?
(c) What are the maintenance indicators for excessive damping?

12. Home Assignment and Follow Up	Prepare a brief note on how damping characteristics influence the overall performance of the AFCS, referencing concepts from LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Gyroscope as an attitude-sensing device, establishment of attitude reference by displacement gyro and rate gyro.	Refer to the master syllabus reference list	LP 23

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min